Syllabus Details

Syllabus ID:	10368
Syllabus Name:	**Data Structures and Algorithm_Cấu trúc dữ liệu và giải thuật**
Course Name English:	**Data Structures and Algorithm**
Subject Code:	**CSD201**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h TE + 85' PE + 102.6h self-study
Pre-Requisite:	PRO192
Description:	Upon finishing the course, students can: 1) Knowledge: Understand (ABET e) - the connection between data structures and their algorithms, including an analysis of algorithms' cor - data structurre in the context of object-oriented program design; - how data structure are implemented in an OO programming language such as Java 2) Able to (ABET e) - organize and manipulate basic structures: array, linked list, tree, heap, hash - use algorithms for traversing, sorting, searching on studying structures - select a suitable algorithm to solve a practical problem 3) Able to (ABET k) - use JAVA programming language for solving some problems - use Eclipse tool for developing programs in JAVA - Implement some programs in JAVA to solve practical problems based on the studying algorithms 4) Others: (ABET i) - Improve study skills (academic reading, information searching, ...)
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examinatic - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	358/QĐ-ĐHFPT dated 04/03/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/3/2024

4 material(s)

No.	Material Description	Au
1	Michaelt T. Goodrich, Roberto Tamassia, Michael H. Goldwasser: Data Structures and Algorithms in Java, 6th Edition, 2014	M
2	FU slides (ppt)	
3	FU exercises (pdf)	
4	Code files for students (java files)	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe the list data structure and its' different way of implementations. Implement the singly linked list. LO1.1 De
2	CLO2	Define stack and queue. Describe basic operations and the use of these structures. LO2.1 Define stack and queue
3	CLO3	Describe about recursive definitions, algorithms, functions and their implementation and use. LO3.1 Describe the b
4	CLO4	Explain about general tree, Binary Tree and Binary Search Tree (BST). Implement BST with basic operations. LO4.1
5	CLO5	Díscuss about graphs and their application. Implement a graph with some basic operations. LO5.1 Demonstrate gr
6	CLO6	Explain the operation and performance of some basic and advanced sorting alggorithms LO6.1 Explain the operati
7	CLO7	Explain about hashing and application LO7.1 Explain the concept of "hash". Define concepts hash function and has
8	CLO8	Describe the Text Processing problem and its' application. Explain the Huffman, LZW and Run-length encoding Al

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials
1	Course Introduction 1.1. Using Arrays 1.2. Singly Linked Lists	Offline	CLO1	IT	Text book, slides, exercises
2	1.3. Circularly Linked Lists 1.4. Doubly Linked Lists	Offline	CLO1	IT	Text book, slides, exercises
3	Progress test and/or Review Exercises	Offline	CLO1	U	Text book, slides, exercises
4	Guiding Exercises/Assignment	Offline	CLO1	U	Text book, slides, exercises
5	2.1 Stacks	Offline	CLO2	ITU	Text book, slides, exercises
6	Guiding Exercises/Assignment	Offline	CLO2	ITU	Text book, slides, exercises
7	2.2 Queues 2.3 Double-Ended Queues (Deque)	Offline	CLO2	T	Text book, slides, exercises
8	2.4 The Priority Queue	Offline	CLO2	T	Text book, slides, exercises
9	Progress test and/or Review Exercises	Offline	CLO2	U	Text book, slides, exercises
10	Guiding Exercises/Assignment	Offline	CLO2	U	Text book, slides, exercises
11	3.1 Illustrative Examples 3.1.1 The Factorial Function 3.1.2 Binary Search 3.1.3 File Systems	Offline	CLO3	IT	Text book, slides, exercises
12	3.2 Analyzing Recursive Algorithms 3.3 Further Examples of Recursion 3.3.1 Linear Recursion 3.3.2 Binary Recursion 3.3.3 Multiple Recursion 3.4 Designing Recursive Algorithms 3.5 Eliminating Tail Recursion	Offline	CLO3	IT	Text book, slides, exercises
13	Progress test and/or Review Exercises	Offline	CLO3	U	Text book, slides, exercises
14	Guiding Exercises/Assignment	Offline	CLO3	U	Text book, slides, exercises

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials
15	4.1 General Trees 4.1.1 Tree Definitions and Properties 4.1.2 The Tree Abstract Data Type	Offline	CLO4	T	Text book, slides, exercises
16	4.2 Binary Trees 4.2.1 The Binary Tree Abstract Data Type 4.2.2 Properties of Binary Trees 4.3 Implementing Trees 4.4 Tree Traversal Algorithms	Offline	CLO4	T	Text book, slides, exercises
17	Progress test and/or Review Exercises Assignment evaluation	Offline	CLO4	U	Text book, slides, exercises
18	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises
19	4.5 Binary Search Trees 4.5.1 Searching Within a Binary Search Tree	Offline	CLO4	T	Text book, slides, exercises
20	4.5.2 Insertions and Deletions	Offline	CLO4	T	Text book, slides, exercises
21	Progress test and/or Review Exercises	Offline	CLO4	U	Text book, slides, exercises
22	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises
23	4.6 Balanced Search Trees 4.7 AVL Trees	Offline	CLO4	T	Text book, slides, exercises
24	4.8 Heaps	Offline	CLO4	T	Text book, slides, exercises
25	Progress test and/or Review Exercises	Offline	CLO4	U	Text book, slides, exercises
26	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises
27	Progress test 1 and review	Offline	CLO4	U	Text book, slides, exercises
28	Review	Offline	CLO4	U	Text book, slides, exercises
29	5.1 Graphs 5.2 Data Structures for Graphs 5.2.1 Edge List Structure 5.2.2 Adjacency List Structure 5.2.3 Adjacency Matrix Structure	Offline	CLO5	IT	Text book, slides, exercises
30	5.3 Graph Traversals 5.3.1 Depth-First Search 5.3.3 Breadth-First Search	Offline	CLO5	IT	Text book, slides, exercises
31	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises
32	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises
33	5.4 Shortest Paths 5.4.1 Weighted Graphs	Offline	CLO5	T	Text book, slides, exercises
34	5.4.2 Dijkstra’s Algorithm	Offline	CLO5	T	Text book, slides, exercises
35	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises
36	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises
37	5.5 Minimum Spanning Trees 5.5.1 Prim-Jarn´ık Algorithm	Offline	CLO5	T	Text book, slides, exercises
38	5.5.2 Kruskal’s Algorithm 5.6. Euler's tour and Euler's cycle	Offline	CLO5	T	Text book, slides, exercises

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials
39	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises
40	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises
41	6.1 Selection-Sort 6.2 Insertion-Sort	Offline	CLO6	IT	Text book, slides, exercises
42	6.3 Bubble-sort 6.4 Quick-Sort	Offline	CLO6	IT	Text book, slides, exercises
43	Progress test and/or Review Exercises	Offline	CLO6	U	Text book, slides, exercises
44	Guiding Exercises/Assignment	Offline	CLO6	U	Text book, slides, exercises
45	6.5 Merge-Sort 6.6 Heap-Sort	Offline	CLO6	T	Text book, slides, exercises
46	6.7 Linear-Time Sorting: Bucket-Sort and Radix-Sort 6.8 Comparing Sorting Algorithms	Offline	CLO6	T	Text book, slides, exercises
47	Progress test and/or Review Exercises	Offline	CLO6	U	Text book, slides, exercises
48	Guiding Exercises/Assignment	Offline	CLO6	U	Text book, slides, exercises
49	7.1 Hash Tables 7.2 Hash Functions 7.3 Collision-Handling	Offline	CLO7	IT	Text book, slides, exercises
50	7.4 Load Factors, Rehashing, and Efficiency 7.5 Java Hash Table Implementation	Offline	CLO7	IT	Text book, slides, exercises
51	Progress test and/or Review Exercises	Offline	CLO7	U	Text book, slides, exercises
52	Guiding Exercises/Assignment	Offline	CLO7	U	Text book, slides, exercises
53	8.1 Abundance of Digitized Text 8.2 Pattern-Matching Algorithms 8.2.1 Brute Force 8.2.2 The Knuth-Morris-Pratt Algorithm	Offline	CLO8	IT	Text book, slides, exercises
54	8.3 Text Compression 8.3.1 The Huffman Coding Algorithm 8.3.2 The LZW Algorithm 8.3.3 The Run-length Encoding Algorithm	Offline	CLO8	IT	Text book, slides, exercises
55	Progress test and/or Review Exercises	Offline	CLO8	U	Text book, slides, exercises
56	Progress test and/or Review Exercises	Offline	CLO8	U	Text book, slides, exercises
57	Assignment evaluation	Offline			
58	Assignment evaluation	Offline			
59	Progress test 2 and review	Offline			
60	Review	Offline			

120 Constructive question(s)

1	1	CQ1.1	Please list the advantages of Linked List over Array?

2	1	HCM_CQ1.1	How do you insert an element into an array?
3	2	CQ1.2	What are the disadvantages of linked lists?
4	2	HCM_CQ1.2	What are the disadvantages of using an array? List some cases that you think using an array is no
5	3	CQ1.3	What are difference between Single, Circularly and Doubly Linked List?
6	3	HCM_CQ1.3	How does a linked list solve the problems of an array? (for inserting, deleting an element)
7	4	CQ2.1	What is a stack?
8	4	HCM_CQ2.1	What is a stack? In your daily lives, describe 3 situations in which the stack principle is used.
9	5	CQ2.2	What are the applications of the stack?
10	5	HCM_CQ2.2	What are the applications of the stack in computer science/information technology?
11	6	CQ2.3	What are the 3 primary methods for a stack?
12	6	HCM_CQ2.3	What are the 3 primary methods for a stack?
13	7	CQ3.1	What is a queue?
14	7	HCM_CQ3.1	What is a queue? What would happen if there is no queue in a coffee shop?
15	8	CQ3.2	What are difference between stack and queue?
16	8	HCM_CQ3.2	What are difference between stack and queue?
17	9	CQ3.3	What is a priority queue?
18	9	HCM_CQ3.3	What is a priority queue? (hint: what happens if an ambulance uses a shared gate with other cars/
19	10	CQ4.1	What is recursive?
20	10	HCM_CQ4.1	What is recursive? What is/are recursive step(s) in Fibonacci number definition?
21	11	CQ4.2	How many main parts does a recursive algorithm have?
22	11	HCM_CQ4.2	How many main parts does a recursive algorithm have?
23	12	CQ4.3	What are pro and cons of recursion?
24	12	HCM_CQ4.3	What are pro and cons of recursion?
25	13	CQ5.1	What are difference between Binary Tree and Binary Search Tree?
26	13	HCM_CQ5.1	What are differences between a Binary Tree and a Binary Search Tree?
27	14	CQ5.2	Which operation of tree is more efficient than array and linked list?
28	14	HCM_CQ5.2	Which operation of tree is more efficient than array and linked list?
29	15	CQ5.3	List the application of BT to specific problems?
30	15	HCM_CQ5.3	List some applications/softwares/functions/systems using BST as the key element?
31	16	CQ6.1	What is an AVL tree?
32	16	HCM_CQ6.1	What will happen if you insert an ordered array into a BST?
33	17	CQ6.2	What are difference between and Binary Search Tree and AVL Tree?
34	17	HCM_CQ6.2	What are different between and Binary Search Tree and AVL Tree?
35	18	CQ6.3	What is the complexity of a AVL tree search?
36	18	HCM_CQ6.3	What is the complexity of a AVL tree for searching? Is an AVL different from a BST when searchin
37	19	CQ7.1	What is heap?
38	19	HCM_CQ7.1	What is a heap? What is the main difference between a min heap and a max heap?
39	20	CQ7.2	What is the similarity and difference between binary tree, AVL tree and heap?

40	20	HCM_CQ7.2	What are the similarities and differences between binary tree, AVL tree and heap?
41	21	CQ7.3	How to calculate balance factor in AVL tree ?
42	21	HCM_CQ7.3	How to calculate balance factor in AVL tree? Show the formula?
43	22	CQ8.1	What is the relationship between the number of edges and the sum of degrees of all vertices in th
44	22	HCM_CQ8.1	What is the relationship between the number of edges and the sum of degrees of all vertices in th
45	23	CQ8.2	What is breadth-first search algorithm?
46	23	HCM_CQ8.2	What is breadth-first search algorithm?
47	24	CQ8.3	What is depth-first search algorithm?
48	24	HCM_CQ8.3	What is depth-first search algorithm? Compare InOrder, PreOrder, PostOrder traversal methods?
49	25	CQ9.1	What is pseudo-graph?
50	25	HCM_CQ9.1	What is a connected-graph?
51	26	CQ9.2	What is a complete graph?
52	26	HCM_CQ9.2	What is a complete graph?
53	27	CQ9.3	What is the formula for calculating the number of edges of a complete graph with n vertices?
54	27	HCM_CQ9.3	What is the formula for calculating the number of edges of a complete graph with n vertices?
55	28	CQ10.1	What is the difference between Dijkstra and Floy algorithms?
56	28	HCM_CQ10.1	What is the difference between Dijkstra and Floy algorithms? (about the input/output and complex
57	29	CQ10.2	How to descript the graph on Java?
58	29	HCM_CQ10.2	How to use the graph structure on Java?
59	30	CQ10.3	What are difference between BFS and DFS?
60	30	HCM_CQ10.3	In a real application, such as GG maps, what happens when a user finds the way from A to B?
61	31	CQ11.1	How to color the graph with the least number of colors?
62	31	HCM_CQ11.1	How to color the graph with the least number of colors?
63	32	CQ11.2	How to find the shortest path between any two vertices?
64	32	HCM_CQ11.2	In a city map, What is the maximum degree of vertices?
65	33	CQ11.3	What is Dijkstra's algorithm?
66	33	HCM_CQ11.3	What is the key idea behind Dijkstra's algorithm?
67	34	CQ12.1	What is the difference between Dijkstra and Floyd algorithms?
68	34	HCM_CQ12.1	Is Floyd algorithm is the combination of n-time Dijkstra's algorithm?
69	35	CQ12.2	How to know if graph has Euler cycle or not?
70	35	HCM_CQ12.2	How to know if a graph has Euler cycle or not?
71	36	CQ12.3	What is a minimum spanning tree?
72	36	HCM_CQ12.3	What is a minimum spanning tree?
73	37	CQ13.1	What are the similarities and differences of Kruskal and Prim Algorithm?
74	37	HCM_CQ13.1	What are the similarities and differences of Kruskal and Prim Algorithm?
75	38	CQ13.2	What is an Euler cycle?
76	38	HCM_CQ13.2	What is different from Euler cycle to Hamilton cycle?
77	39	CQ13.3	What is Hamilton cycle?

78	39	HCM_CQ13.3	Are there any methods/algorithms to determine if a graph has Hamilton?
79	40	CQ14.1	What is the complexity of the select, insert and bubble algorithm?
80	40	HCM_CQ14.1	What is the complexity of the selection, insertion and bubble sort algorithms? How do you find yo
81	41	CQ14.2	Which of the 3 algorithms for selection, insertion and bubble is faster? Why?
82	41	HCM_CQ14.2	Which of the 3 algorithms for selection, insertion and bubble is faster? Why?
83	42	CQ14.3	Which sorting algorithms do you know of, which can handle multithreading?
84	42	HCM_CQ14.3	Among the sorting algorithms you know, which can handle multithread programming?
85	43	CQ15.1	What is Quick sort?
86	43	HCM_CQ15.1	What is Quick sort? What is the key idea behind this algorithm?
87	44	CQ15.2	Can you list the names of the sorting algorithms that belong to the divide-and-conquer algorithm?
88	44	HCM_CQ15.2	Can you list the names of the sorting algorithms that belong to the divide-and-conquer algorithm?
89	45	CQ15.3	Which sorting algorithm needs more space to complete the classification?
90	45	HCM_CQ15.3	Which sorting algorithm needs more space to complete the classification?
91	46	CQ16.1	What is Merge sort?
92	46	HCM_CQ16.1	What is Merge sort? How is this algorithm different from Quick sort?
93	47	CQ16.2	Which algorithm for sorting is the fastest? Why? (Selection, Insertion, Bubble, Quick, Merge, Heap
94	47	HCM_CQ16.2	What are the main weaknesses of radix sort?
95	48	CQ16.3	Which sorting algorithm is best for memory? Why? (Selection, Insertion, Bubble, Quick, Merge, He
96	48	HCM_CQ16.3	Which sorting algorithm is best for memory?
97	49	CQ17.1	What are the advantages and disadvantages of hashing?
98	49	HCM_CQ17.1	What are the advantages and disadvantages of hashing?
99	50	CQ17.2	How to select hash functions?
100	50	HCM_CQ17.2	How to select hash functions? List some principles to select/design a good hash function?
101	51	CQ17.3	What is a perfect hash function?
102	51	HCM_CQ17.3	What is a perfect hash function?
103	52	CQ18.1	How many ways to resolve collisions in hashing?
104	52	HCM_CQ18.1	How many ways to resolve collisions in hashing? Compare them in term of time and memory com
105	53	CQ18.2	What is the complexity of Brute-Force algorithm?
106	53	HCM_CQ18.2	What is the complexity of Brute-Force algorithm in text pattern matching?
107	54	CQ18.3	What is the complexity of Knuth-Morris-Pratt algorithm?
108	54	HCM_CQ18.3	What is the complexity of Knuth-Morris-Pratt algorithm? What is the role of T[i] array?
109	55	CQ19.1	What Is the difference in lossy vs lossless compression?
110	55	HCM_CQ19.1	What Is the difference between lossy and lossless compression?
111	56	CQ19.2	How to compare the efficiency of different data compression methods?
112	56	HCM_CQ19.2	How to compare the efficiency of different data compression methods?
113	57	CQ19.3	What is Huffman encoding?
114	57	HCM_CQ19.3	What is Huffman encoding? What is the key idea of this method?
115	58	CQ20.1	What is LZW encoding ?

116	58	HCM_CQ20.1	What is LZW encoding ? How does this method differ from Huffman method?
117	59	CQ20.2	What is Run-Length encoding ?
118	59	HCM_CQ20.2	What is Run-Length encoding ? Do you think this algorithm is used to compress the cartoon movie
119	60	CQ20.3	What is the difference between Huffman and LZW coding techniques?
120	60	HCM_CQ20.3	List some lossy compression methods? Explain why they are used even they cause information lo

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration
1	Progress test (PT)	quiz	2	20.0%	>0	Option 1: 20'-40' Option 2 (For Constructivism Appro
2	Assignment (AS)	on-going	2	20.0%	>0	Option 1: at home Option 2 (For Constructivism Appro
3	Practical Exam	practical exam	1	30.0%	>0	85'
4	Final exam	Final exam	1	30.0%	4	60'